

# Devoir de BSM du 28 Juillet 2026

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## Abstract

*Dans ce travail, il nous a été demandé de calculer l'amplitude  $\mathcal{A}(1^+, 2^+, 3^+, 4^-, 5^-, 6^-)$  en utilisant la technique de récursion BCFW:*

$$\mathcal{A}(1\dots n) = \sum \mathcal{A}_L \frac{1}{p_m^2} \mathcal{A}_R$$

## Rappels sur les formules de Parke-Taylor et formules des amplitudes $\mathcal{A}_3$

### Parke-Taylor

$$\begin{aligned} \mathcal{A}_{MHV}(1^+, \dots, i^-, \dots, j^-, \dots, n^+) &= \frac{\langle ij \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle} \\ \mathcal{A}_{anti-MHV}(1^-, \dots, i^+, \dots, j^+, \dots, n^-) &= \frac{[ij]^4}{[12][23] \dots [n1]} \end{aligned} \tag{1}$$

### $\mathcal{A}_3$

$$\begin{aligned} \mathcal{A}_3(1^+, 2^-, 3^-) &= \frac{\langle 23 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 31 \rangle} \\ \mathcal{A}_3(1^-, 2^+, 3^+) &= \frac{[23]^4}{[12][23][31]} \end{aligned} \tag{2}$$

## 1 Choix de shift

Dans ce qui suit, nous avons utilisé le shift [61]:

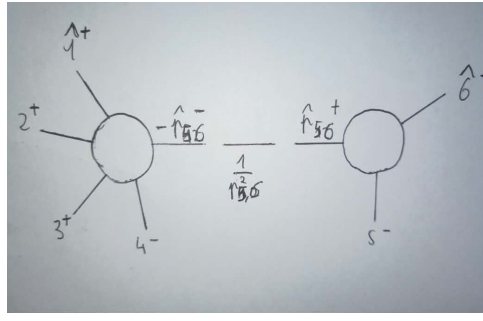
- $\hat{\bar{\lambda}}_6 = \bar{\lambda}_6 + z\bar{\lambda}_1$
- $\hat{\lambda}_1 = \lambda_1 - z\lambda_6$

## 2 Diagrammes possibles

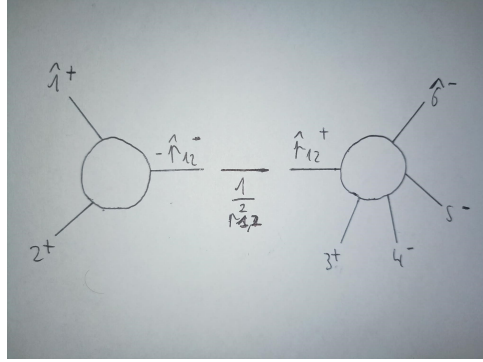
Avec notre choix de shift et le choix d'hélicité des impulsions  $\hat{p}_m$  pour tenir compte du propagateur, nous avons uniquement deux décompositions d'amplitude qui ne valent pas zéros.

Nous les notons  $\mathbb{X}$  et  $\mathbb{Y}$ , voici leurs représentations graphiques:

Pour  $\mathbb{X}$



Pour  $\mathbb{Y}$



### 3 L'amplitude $\mathbb{X} = \mathcal{A}(\hat{1}^+, 2^+, 3^+, 4^-, -\hat{p}_{5,6}^-) \frac{1}{p_{5,6}^2} \mathcal{A}(\hat{p}_{5,6}^+, 5^-, 6^-)$

Nous partons de:

$$\mathbb{X} = \mathcal{A}(\hat{1}^+, 2^+, 3^+, 4^-, -\hat{p}_{5,6}^-) \frac{1}{p_{5,6}^2} \mathcal{A}(\hat{p}_{5,6}^+, 5^-, 6^-)$$

En utilisant la formule de Parke-Taylor et des amplitudes  $\mathcal{A}_3$  pour les MHV et en tenant compte du shift que l'on a choisit, cette expression devient:

$$\begin{aligned} \mathbb{X} &= \frac{(-1)^4 \langle 4\hat{p}_{5,6} \rangle^4}{\langle \hat{1}2 \rangle \langle 23 \rangle \langle 34 \rangle (-1) \langle 4\hat{p}_{5,6} \rangle (-1) \langle \hat{p}_{5,6} \hat{1} \rangle} \times \frac{1}{p_{5,6}^2} \times \frac{\langle 56 \rangle^4}{\langle \hat{p}_{5,6} 5 \rangle \langle 56 \rangle \langle 6\hat{p}_{5,6} \rangle} \\ &= \frac{\langle 4\hat{p}_{5,6} \rangle^3}{\langle \hat{1}2 \rangle \langle 23 \rangle \langle 34 \rangle \langle \hat{p}_{5,6} \hat{1} \rangle} \times \frac{1}{p_{5,6}^2} \times \frac{\langle 56 \rangle^3}{\langle \hat{p}_{5,6} 5 \rangle \langle 6\hat{p}_{5,6} \rangle} \\ &= \frac{\langle 4\hat{p}_{5,6} \rangle^3 [\hat{p}_{5,6} 1]^3}{\langle \hat{1}2 \rangle \langle 23 \rangle \langle 34 \rangle \langle \hat{p}_{5,6} \hat{1} \rangle [\hat{p}_{5,6} 1]} \times \frac{1}{p_{5,6}^2} \times \frac{\langle 56 \rangle^3}{\langle \hat{p}_{5,6} 5 \rangle [\hat{p}_{5,6} 1] \langle 6\hat{p}_{5,6} \rangle [\hat{p}_{5,6} 1]} \end{aligned} \quad (3)$$

Il nous faut maintenant calculer certains termes dans cette expression.

#### 3.1

$$\begin{aligned} \langle 4\hat{p}_{5,6} \rangle [\hat{p}_{5,6} 1] &= \langle 4(\hat{p}_{5,6}) 1 \rangle \\ &= \langle 4(k_5 + \hat{k}_6) 1 \rangle \\ &= \langle 4(\lambda_5 \bar{\lambda}_5 + \lambda_6 \hat{\lambda}_6) 1 \rangle \\ &= \langle 4\lambda_5 \bar{\lambda}_5 1 \rangle + \langle 4\lambda_6 \hat{\lambda}_6 1 \rangle \\ &= \langle 451 \rangle + \langle 4\hat{6}1 \rangle \\ &= \langle 451 \rangle + \langle 46 \rangle [\hat{6}1] \\ &= \langle 451 \rangle + \langle 46 \rangle ([61] + z[11]) \\ &= \langle 4(5+6)1 \rangle \end{aligned} \quad (4)$$

### 3.2

$$\begin{aligned}
\langle \hat{p}_{5,6} \hat{1} \rangle [\hat{p}_{5,6} 1] &= -\langle \hat{1} \hat{p}_{5,6} \rangle [\hat{p}_{5,6} 1] \\
&= -(\langle \hat{1} 5 \rangle [51] + \langle \hat{1} 6 \rangle [\hat{6} 1]) \\
&= -(\langle 15 \rangle - z \langle 65 \rangle) [51] - (\langle 16 \rangle - z \langle 66 \rangle) ([61] + z [11]) \\
&= -\langle 15 \rangle [51] - \langle 16 \rangle [61] + z \langle 65 \rangle [51]
\end{aligned} \tag{5}$$

Or, on a:

$$k_5 + \hat{k}_6 + \hat{p}_{5,6} = 0 \Leftrightarrow \hat{p}_{5,6} = -(k_5 + k_6)$$

Ce qui nous donne:

$$\hat{p}_{5,6}^2 = 0 \Leftrightarrow 2k_5 \hat{k}_6 = \langle 56 \rangle [\hat{6} 5] = 0$$

En développant, on trouve que:

$$\langle 56 \rangle ([65] + z [15]) = 0 \Leftrightarrow z = -\frac{[65]}{15} = -\frac{[56]}{[51]} = \frac{[65]}{[51]}$$

En injectant cette valeur de z dans la dernière forme de (5), on trouve que:

$$\langle \hat{p}_{5,6} \hat{1} \rangle [\hat{p}_{5,6} 1] = -\langle 15 \rangle [51] - \langle 16 \rangle [61] - \langle 65 \rangle [56]$$

Or, on a aussi que:

$$(k_{5,1})^2 = (k_5 + k_6 + k_1)^2 = 2k_5 k_6 + 2k_5 k_1 + 2k_6 k_1 = \langle 15 \rangle [51] + \langle 16 \rangle [61] + \langle 65 \rangle [56]$$

Ainsi, nous obtenons l'expression:

$$\langle \hat{p}_{5,6} \hat{1} \rangle [\hat{p}_{5,6} 1] = -(k_{5,1})^2$$

### 3.3

$$\begin{aligned}
\langle \hat{p}_{5,6} 5 \rangle [\hat{p}_{5,6} 1] &= -\langle 5 \hat{p}_{5,6} \rangle [\hat{p}_{5,6} 1] \\
&= -\langle 55 \rangle [51] - \langle 56 \rangle [\hat{6} 1] \\
&= -\langle 56 \rangle ([61] + z [11]) \\
&= -\langle 56 \rangle [61]
\end{aligned} \tag{6}$$

### 3.4

$$\begin{aligned}
\langle 6 \hat{p}_{5,6} \rangle [\hat{p}_{5,6} 1] &= \langle 65 \rangle [51] + \langle 66 \rangle [\hat{6} 1] \\
&= \langle 65 \rangle [51]
\end{aligned} \tag{7}$$

### 3.5

$$\begin{aligned}
p_{5,6}^2 &= 2k_5 k_6 \\
&= \langle 56 \rangle [65]
\end{aligned} \tag{8}$$

### 3.6

$$\begin{aligned}
\langle \hat{1}2 \rangle &= \langle 12 \rangle - z \langle 62 \rangle \\
&= \langle 12 \rangle - \frac{[65]}{[51]} \langle 62 \rangle \\
&= \frac{\langle 12 \rangle [51] - \langle 62 \rangle [65]}{[51]} \\
&= \frac{\langle 21 \rangle [15] + \langle 26 \rangle [65]}{[51]} \\
&= \frac{\langle 2(1+6)5 \rangle}{[51]}
\end{aligned} \tag{9}$$

### 3.7 L'expression finale de l'amplitude $\mathbb{X}$

On avait:

$$\mathbb{X} = \frac{\langle 4\hat{p}_{5,6} \rangle^3 [\hat{p}_{5,6}1]^3}{\langle \hat{1}2 \rangle \langle 23 \rangle \langle 34 \rangle \langle \hat{p}_{5,6}\hat{1} \rangle [\hat{p}_{5,6}1]} \times \frac{1}{p_{5,6}^2} \times \frac{\langle 56 \rangle^3}{\langle \hat{p}_{5,6}5 \rangle [\hat{p}_{5,6}1] \langle 6\hat{p}_{5,6} \rangle [\hat{p}_{5,6}1]}$$

En injectant les expressions trouvées de 3.1 à 3.6 dans cette dernière, on trouve:

$$\begin{aligned}
\mathbb{X} &= \frac{\langle 4(5+6)1 \rangle^3 \langle 56 \rangle^3}{\frac{\langle 2(1+6)5 \rangle}{[51]} \langle 23 \rangle \langle 34 \rangle (-k_{5,1})^2 \langle 56 \rangle [65] (-1) \langle 56 \rangle [61] \langle 65 \rangle [51]} \\
&= \frac{\langle 4(5+6)1 \rangle^3 \langle 56 \rangle^3}{\frac{\langle 2(1+6)5 \rangle}{[51]} \langle 23 \rangle \langle 34 \rangle (-k_{5,1})^2 \langle 56 \rangle (-1) [56] (-1) \langle 56 \rangle [61] (-1) \langle 56 \rangle [51]} \\
\mathbb{X} &= \frac{\langle 4(5+6)1 \rangle^3}{(k_{5,1})^2 \langle 23 \rangle \langle 34 \rangle [56] [61] \langle 2(1+6)5 \rangle}
\end{aligned} \tag{10}$$

## 4 L'amplitude $\mathbb{Y} = \mathcal{A}(\hat{1}^+, 2^+, -\hat{p}_{1,2}^-, \frac{1}{p_{1,2}^2} \mathcal{A}(\hat{p}_{1,2}^+, 3^+, 4^-, 5^-, 6^-))$

Nous partons cette fois de l'expression suivante:

$$\mathbb{Y} = \mathcal{A}(\hat{1}^+, 2^+, -\hat{p}_{1,2}^-) \frac{1}{p_{1,2}^2} \mathcal{A}(\hat{p}_{1,2}^+, 3^+, 4^-, 5^-, 6^-)$$

En utilisant cette fois-ci les formules de Parke-Taylor et  $\mathcal{A}$  pour les amplitudes anti-MHV, on trouve que:

$$\begin{aligned}
\mathbb{Y} &= \frac{[12]^4}{[12](-1)[2\hat{p}_{1,2}](-1)[\hat{p}_{1,2}1]} \times \frac{1}{p_{1,2}^2} \times \frac{[\hat{p}_{1,2}3]^4}{[\hat{p}_{1,2}3][34][45][5\hat{6}][\hat{6}\hat{p}_{1,2}]} \\
&= \frac{[12]^3}{[2\hat{p}_{1,2}][\hat{p}_{1,2}1]} \times \frac{1}{p_{1,2}^2} \times \frac{[\hat{p}_{1,2}3]^3}{[34][45][5\hat{6}][\hat{6}\hat{p}_{1,2}]} \\
&= \frac{[12]^3}{\langle 6\hat{p}_{1,2} \rangle [2\hat{p}_{1,2}] \langle 6\hat{p}_{1,2} \rangle [\hat{p}_{1,2}1]} \times \frac{1}{p_{1,2}^2} \times \frac{\langle 6\hat{p}_{1,2} \rangle^3 [\hat{p}_{1,2}3]^3}{[34][45][5\hat{6}] \langle 6\hat{p}_{1,2} \rangle [\hat{6}\hat{p}_{1,2}]}
\end{aligned} \tag{11}$$

Nous allons maintenant calculer les termes petit à petit pour simplifier cette expression

#### 4.1

$$\begin{aligned}
\langle 6\hat{p}_{1,2} \rangle [\hat{p}_{1,2}3] &= \langle 6\hat{1} \rangle [13] + \langle 62 \rangle [23] \\
&= (\langle 61 \rangle - z \langle 66 \rangle) [13] + \langle 62 \rangle [23] \\
&= \langle 61 \rangle [13] + \langle 62 \rangle [23] \\
&= \langle 6(1+2)3 \rangle
\end{aligned} \tag{12}$$

## 4.2

$$\begin{aligned}
\langle 6\hat{p}_{1,2} \rangle [2\hat{p}_{1,2}] &= -\langle 6\hat{p}_{1,2} \rangle [\hat{p}_{1,2}2] \\
&= -(\langle 6\hat{1} \rangle [12] + \langle 62 \rangle [22]) \\
&= (\langle 61 \rangle - z\langle 66 \rangle) [12] \\
&= -\langle 61 \rangle [12]
\end{aligned} \tag{13}$$

## 4.3

$$\begin{aligned}
\langle 6\hat{p}_{1,2} \rangle [\hat{p}_{1,2}1] &= \langle 6\hat{1} \rangle [11] + \langle 62 \rangle [21] \\
&= \langle 62 \rangle [21]
\end{aligned} \tag{14}$$

## 4.4

$$\begin{aligned}
\langle 6\hat{p}_{1,2} \rangle [\hat{6}\hat{p}_{1,2}] &= -\langle 6\hat{p}_{1,2} \rangle [\hat{p}_{1,2}\hat{6}] \\
&= -(\langle 6\hat{1} \rangle [1\hat{6}] + \langle 62 \rangle [2\hat{6}]) \\
&= -(\langle 61 \rangle - z\langle 66 \rangle)([16] + z[11]) - \langle 62 \rangle ([26] + z[21]) \\
&= -\langle 61 \rangle [16] - \langle 62 \rangle [26] - z\langle 62 \rangle [21]
\end{aligned} \tag{15}$$

Or, nous savons que :  $-\hat{p}_{1,2} + \hat{k}_1 + k_2 = 0 \Leftrightarrow \hat{p}_{1,2} = \hat{k}_1 + k_2$

Ainsi :  $(\hat{p}_{1,2})^2 = 2\hat{k}_1 k_2 = \langle \hat{1}2 \rangle [21] = 0$

Ce qui nous donne en développant le shift:  $(\langle 12 \rangle - z\langle 62 \rangle)[21] = 0 \Leftrightarrow \langle 12 \rangle - z\langle 62 \rangle = 0 \Leftrightarrow z = \frac{\langle 12 \rangle}{\langle 62 \rangle}$

En injectant dans (15), on trouve que:

$$\langle 6\hat{p}_{1,2} \rangle [\hat{6}\hat{p}_{1,2}] = -\langle 61 \rangle [16] - \langle 62 \rangle [26] - \langle 12 \rangle [21]$$

Or, nous avons aussi :

$$\begin{aligned}
k_{6,2} = k_6 + k_1 + k_2 &\Rightarrow (k_{6,2})^2 = 2k_6 k_1 + 2k_1 k_2 + 2k_6 k_2 \\
&= \langle 61 \rangle [16] + \langle 62 \rangle [26] + \langle 12 \rangle [21]
\end{aligned} \tag{16}$$

Par conséquent, nous pouvons faire l'identification suivante:

$$\langle 6\hat{p}_{1,2} \rangle [\hat{6}\hat{p}_{1,2}] = -(k_{6,2})^2$$

## 4.5

$$\begin{aligned}
(p_{1,2})^2 &= (k_1 + k_2)^2 \\
&= 2k_1 k_2 \\
&= \langle 12 \rangle [21]
\end{aligned} \tag{17}$$

## 4.6

$$\begin{aligned}
[5\hat{6}] &= [56] + z[51] \\
&= [56] + \frac{\langle 12 \rangle}{\langle 62 \rangle} [51] \\
&= \frac{[56]\langle 62 \rangle + \langle 12 \rangle [51]}{\langle 62 \rangle} \\
&= \frac{\langle 26 \rangle [65] + \langle 21 \rangle [15]}{\langle 62 \rangle} \\
&= \frac{\langle 2(1+6)5 \rangle}{\langle 62 \rangle}
\end{aligned} \tag{18}$$

## 4.7 L'expression finale de l'amplitude $\mathbb{Y}$

On avait:

$$\mathbb{Y} = \frac{[12]^3}{\langle 6\hat{p}_{1,2} \rangle [2\hat{p}_{1,2}] \langle 6\hat{p}_{1,2} \rangle [\hat{p}_{1,2}1]} \times \frac{1}{p_{1,2}^2} \times \frac{\langle 6\hat{p}_{1,2} \rangle^3 [\hat{p}_{1,2}3]^3}{[34][45][5\hat{6}] \langle 6\hat{p}_{1,2} \rangle [\hat{6}\hat{p}_{1,2}]}$$

En injectant les expressions trouvées de 4.1 à 4.6 dans cette dernière, on trouve que:

$$\begin{aligned} \mathbb{Y} &= \frac{[12]^3}{(-1)\langle 61 \rangle [12] \langle 62 \rangle [21]} \times \frac{1}{\langle 12 \rangle [21]} \times \frac{\langle 6(1+2)3 \rangle^3}{[34][45] \frac{\langle 2(1+6)5 \rangle}{\langle 62 \rangle} (-1)(k_{6,2})^2} \\ &= \frac{[12]^3}{(-1)\langle 61 \rangle [12] \langle 62 \rangle (-1)[12]} \times \frac{1}{\langle 12 \rangle (-1)[12]} \times \frac{\langle 6(1+2)3 \rangle^3}{[34][45] \frac{\langle 2(1+6)5 \rangle}{\langle 62 \rangle} (-1)(k_{6,2})^2} \\ \mathbb{Y} &= \frac{\langle 6(1+2)3 \rangle^3}{(k_{6,2})^2 \langle 61 \rangle \langle 12 \rangle [34][45] \langle 2(1+6)5 \rangle} \end{aligned} \quad (19)$$

## 5 L'amplitude totale $\mathcal{A}(1^+, 2^+, 3^+, 4^-, 5^-, 6^-)$

On a :

$$\mathcal{A}(1^+, 2^+, 3^+, 4^-, 5^-, 6^-) = \mathbb{X} + \mathbb{Y}$$

Et on a :

$$\begin{aligned} \mathbb{X} &= \frac{\langle 4(5+6)1 \rangle^3}{(k_{5,1})^2 \langle 23 \rangle \langle 34 \rangle [56][61] \langle 2(1+6)5 \rangle} \\ \mathbb{Y} &= \frac{\langle 6(1+2)3 \rangle^3}{(k_{6,2})^2 \langle 61 \rangle \langle 12 \rangle [34][45] \langle 2(1+6)5 \rangle} \end{aligned} \quad (20)$$

Et en faisant la somme, on trouve finalement que:

$$\mathcal{A}(1^+, 2^+, 3^+, 4^-, 5^-, 6^-) = \frac{\langle 4(5+6)1 \rangle^3}{(k_{5,1})^2 \langle 23 \rangle \langle 34 \rangle [56][61] \langle 2(1+6)5 \rangle} + \frac{\langle 6(1+2)3 \rangle^3}{(k_{6,2})^2 \langle 61 \rangle \langle 12 \rangle [34][45] \langle 2(1+6)5 \rangle}$$